

FPGA/HLS Brain-Wave Recognizer

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Abstract - This report presents the design, implementation, and verification of a real-time EEG-like band recognizer implemented on the Nexys 4 DDR FPGA board. The project combines a Vitis HLS response band detector with VHDL modules for synthetic sample generation, VGA visualization, waveform buffering, and PWM audio sonification. The design operates from a 100 MHz board clock and a 25 MHz internal clock generated by a Clocking Wizard, while the logical signal-processing sample rate is 128 Hz. Five selectable bands are supported: Delta, Theta, Alpha, Beta, and Gamma.

Keywords – FPGA, VHDL, Vitis HLS, FIR filter, EEG bands, VGA, PWM audio, Nexys 4 DDR.

Formatting and citations follow general IEEE style with a change to 12-point font size of normal text and headings, as well as paragraph starts indentation for the ease of reading.

I. INTRODUCTION

A. Background

The objective of this project was to design an FPGA-based signal recognizer that is capable of detecting EEG-like frequency bands and exposing the result through both visual and audio outputs. The system is educational rather than medical: the input is a synthetic signal generated inside the FPGA, not a real analog EEG front-end.

The final design follows a stream-like architecture:

1. VHDL test signal generator produces signed Q2.14 samples and a one-clock valid sample flag pulse.
2. HLS recognizer consumes only valid samples, filters them with a selectable FIR band-pass filter, extracts an envelope, compares it against a threshold, and produces a detection flag.
3. VGA dashboard shows the result, which is tonified using PWM audio.

The project repository with further comments, proof of validation and essential files is available publicly on Github [1].

B. Motivation

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C. Context

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II. OBJECTIVES AND SPECIFICATION

FPGA demonstrator was built around an HLS signal-processing core. The project had to be observable without a software terminal, thus

VGA and PWM outputs were added to make the recognition result visible and audible.

Table 1: Main design specifications

Item	Value
Board	Nexys 4 DDR
FPGA part	xc7a100tcsq324-1
Toolchain	Vitis HLS / Vivado 2024.1
Board clock	100 MHz
Internal clock	25 MHz
Sample rate	128 samples/s
Input source	Synthetic sine generator
Main outputs	VGA and PWM audio

The recognizer supports five EEG-like frequency bands. The 128 Hz sample rate gives a 64 Hz Nyquist frequency, so the Gamma band is limited to 30-62 Hz to remain below the sampling limit.

Table 2: Supported EEG-Like Bands

Selector	Band	Test Frequency	Approximate Range
0	Delta	2 Hz	0.5-4 Hz
1	Theta	6 Hz	4-8 Hz
2	Alpha	10 Hz	8-12 Hz
3	Beta	20 Hz	12-30 Hz
4	Gamma	40 Hz	30-62 Hz

III. SYSTEM ARCHITECTURE

The complete system is composed of four main functional blocks:

- **Test Signal Generator:** produces synthetic signed Q2.14 sine samples and a one-clock *sample_valid* pulse at 128 Hz.
- **HLS Brain-Wave Recognizer:** filters the input sample using one of five FIR coefficient sets, computes an envelope, applies a threshold, and produces a detection flag.
- **VGA Display:** generates 640 x 480 VGA timing and draws the dashboard, including the detection banner, envelope bar, threshold marker, and waveform plot.

- **PWM Audio:** generates a PWM tone when detection is active, with the tone frequency selected by the detected band.

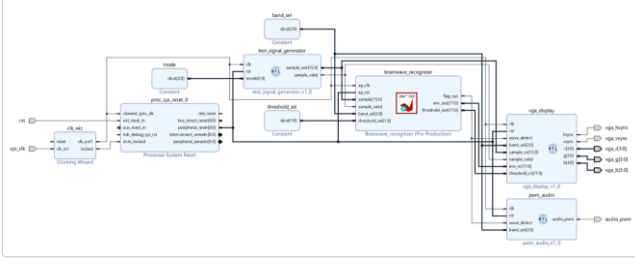


Figure 1: Top-level architecture of the Brain-Wave Recognizer project [1].

Sample signal and *sample_valid* pulse are shared by the HLS detector and VGA waveform buffer. HLS outputs *flag_out*, *env_out*, and *threshold_out* drive VGA and PWM outputs.

Table 3: Main inter-module signals.

Signal	Purpose
sample	Signed Q2.14 input sample
sample_valid	One-clock update pulse
band_sel	Band selection 0-4
threshold_sel	Low/mid/high threshold
flag_out	Sustained detection flag
env_out	Envelope for VGA bar
audio_pwm	PWM audio output

IV. COMPONENTS

A. HLS Brain-Wave Recognizer

The main component is *brainwave_recognizer*, implemented in Vitis HLS. The detector stores the most recent 32 valid samples in a shift register. When *sample_valid* is true, the new sample is inserted into *x_reg[0]*, the selected coefficient row is read using *band_sel*, and a 32-tap multiply-accumulate operation computes the FIR output. If *sample_valid* is false, the shift register, envelope, and persistence counter are left unchanged.

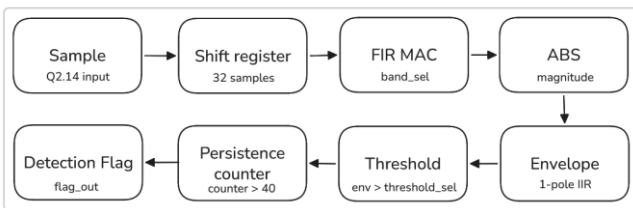


Figure 2: Internal processing chain of the HLS detector.

B. Test Signal Generator

The VHDL *test_signal_generator* module creates synthetic EEG-like sine waves. It contains a 128-entry sine lookup table with half-scale Q2.14 values (a phase step of 10 produces a 10 Hz Alpha test signal).

A fractional accumulator generates *sample_valid* at exactly 128 Hz on average while the hardware clock runs at 25 MHz. Since $25,000,000 / 128 = 195,312.5$, the interval alternates between 195,313 and 195,312 clock cycles. This avoids the error that would result from using a fixed integer divider.

C. VGA Dashboard

VGA module generates 640x480 dashboard with a 25 MHz pixel clock. It drives active-low hsync and vsync and blanks RGB outside the visible region. Inside the visible region, a graphical dashboard is drawn: a top banner indicates detection state and selected band, an envelope bar shows current signal strength, a red marker shows the threshold, and a waveform plot displays recent input samples.

Each valid input sample is converted from signed Q2.14 to a pixel coordinate inside the waveform plot. At 128 samples/s, the 640-point buffer represents approximately five seconds of signal history.

D. PWM Audio Output

The *pwm_audio* module provides an audible indication when a band is detected. It maps the selected EEG band to an audible tone: Delta to about 262 Hz, Theta to 330 Hz, Alpha to 440 Hz, Beta to 660 Hz, and Gamma to 880 Hz. These are demo tones, not EEG frequencies.

V. VERIFICATION

A. HLS Verification

HLS testbench verifies Alpha detection, *sample_valid* hold behavior, rejection of a Beta-frequency into the Alpha filter, positive detection for all five supported bands, variable clamping, and threshold-level behavior. The C simulation reports that all HLS tests passed, and C/RTL co-simulation reports VHDL PASS.

Table 4: Csynth results [1].

Metric	Result
Target clock	10.00 ns
Estimated clock	6.940 ns
Latency	8 cycles
Interval	1 cycle

BRAM_18K	0
DSP	32
FF	2343
LUT	942

```

Starting HLS testbench...
TEST: alpha positive detection
PASS: alpha positive detection
TEST: sample_valid=false hold behavior
PASS: sample_valid=false hold behavior
TEST: 20 Hz beta into alpha filter should not detect
PASS: beta rejected by alpha filter
TEST: all-band positive detection
band=0 freq=2.000000 env=51528 threshold=23592 flag=1
band=1 freq=6.000000 env=43002 threshold=23592 flag=1
band=2 freq=10.000000 env=42262 threshold=23592 flag=1
band=3 freq=20.000000 env=41892 threshold=23592 flag=1
band=4 freq=40.000000 env=42201 threshold=23592 flag=1
PASS: all-band positive detection
TEST: invalid selector clamping
PASS: invalid selector clamping
TEST: threshold levels
low threshold: env=21132 threshold=13104 flag=1
high threshold: env=21132 threshold=32768 flag=0
PASS: threshold levels
All HLS tests passed.

```

Figure 3: HLS verification result showing passing simulation/co-simulation [1].

B. RTL Simulation

Separate RTL testbenches were used for the signal generator, VGA display, and PWM audio block:

- Signal-generator testbench verifies the real 25 MHz / 128 Hz timing and checks Alpha and silence modes.
- VGA testbench verifies interface consistency, sync timing, blanking, and detection-banner behavior.
- PWM testbench verifies activity when detection is active and inactive, all valid band selectors, and invalid-selector default behavior.

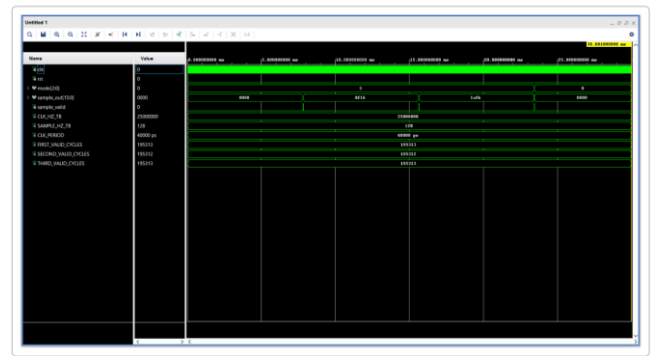


Figure 4: RTL simulation waveform for signal generator [1].

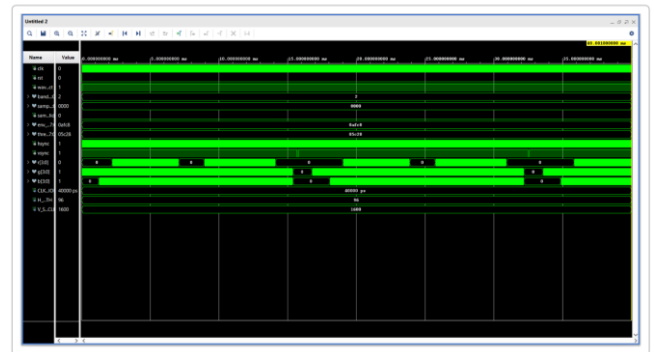


Figure 5: RTL simulation waveform for VGA [1].

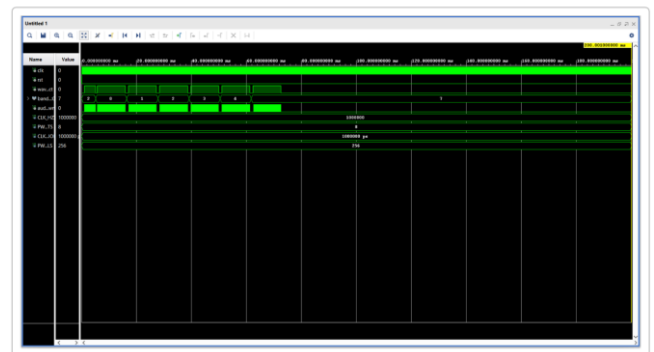


Figure 6: RTL simulation waveform for PWM [1].

C. Vivado Implementation

After HLS IP export, the complete system was integrated in Vivado and implemented for the Nexys 4 DDR FPGA. The post-implementation timing report shows that all user specified timing constraints are met. The most relevant implementation figures are summarized below.

Table 5: Post-implementation utilization, timing, and power summary [1].

Resource	Used	Util.
Slice LUTs	3886	6.13%
Slice registers	6893	5.44%
DSP blocks	34	14.17%
Block RAM	0	0.00%
Bonded IOB	17	8.10%
Total power	0.242 W	-

WNS	15.571 ns	pass
WHS	0.033 ns	pass

<https://docs.amd.com/r/2024.1-English/ug910-vivado-getting-started/Documentation-Suite>.
[Accessed 25 June 2026].

DRC report contains warnings of optimization suggestions about additional pipelining, but no functional errors.

D. Board-Level Demonstration

The intended board demonstration uses the Alpha configuration: mode = 3, band_sel = 2, and threshold_sel = 1. In this configuration, the synthetic generator produces a 10 Hz Alpha-like sine wave, HLS detector selects the Alpha FIR coefficients, and the medium threshold is used. The board showcases a green VGA status banner, an envelope bar above the threshold marker, a visible waveform trace, and audible PWM output while detection is active.

TODO ADD IMAGE AND EXPLANATION
IF DONE ON SIGHT

VI. LIMITATIONS

The project has several limitations, as it uses synthetic EEG-like input and is not a medical EEG device. As well as has no analog front-end, ADC interface, electrode safety circuitry, or real biomedical validation.

VII. CREDITS

The laboratory report was based on the course project slides. The project was developed using AMD/Xilinx Vitis HLS [2] and Vivado documentation [3] and the Nexys 4 DDR board constraints [1]. AI agents were used to help draft comments, markdown descriptions, and report text, while the code and generated artifacts were checked and revised by the authors.

VIII. References

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